

Table 2.5 Cobalt Chrome Alloys [20], [23], [25], [36].

ASTM (UNS)	Chemical Composition	Thermal Treatment Condition	Elastic Modulus [GPa]	Yield Strength [MPa]	Tensile Strength [MPa]
F75-12 (R30075)	Co 28Cr 6Mo	As-cast	210-250	450	655
		Annealed	210-250	517	890
F90-14 (R30605)	Co 20Cr 15W 10Ni	Annealed ^a (Bar & Wire)	210	310	860
		Cold-Worked ^a (Bar & Wire)	210	760	1250
		Annealed ^b (Sheet & Strip)	210	379	896
F562-13 (R30035)	35Co 35Ni 20Cr 10Mo	Annealed	230	241-586	793-1069
		Cold-Worked	230	655-1586	1000-1793
F688-14 (R30035)	Co 35Ni 20Cr 10Mo	Annealed (Sheet)	230	310	792
		Cold-Worked (48%)	230	1343	1357
F799-11 (R31537- 38-39)	Co 28Cr 6Mo	Forgings	210-230	827	1172
F961-14 (R30035)	35Co 35Ni 20Cr 10Mo	Forgings ^c	230	*	*
F1537-11 (R31537- 38-39)	Co 28Cr 6Mo	Annealed	210	517	897
		Hot Worked	230	700	1000
		Warm Worked	230	827	1172
F2886-10 (R30075)	Co 28Cr 6Mo	Low Carbon	210-250	450	725
		High Carbon	210-250	480	825

a: bar and wire, b: sheet and strip, c: comply with the minimum mechanical properties as specified in Specification F562